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Valuing Diversity: CEOs' Career Experiences and Corporate Investment

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Abstract. This paper investigates the impact of CEOs' career experiences on corporate investment decisions. We hypothesize that CEOs with more diverse career experiences are less likely to be constrained by insufficient internal capital. The potential mechanism is that rich external experiences help CEOs accumulate social connections and these connections mitigate information asymmetry and lead to better access to external funds. Consistent with this argument, we find that firms with CEOs who have more diverse career experiences exhibit lower investment-cash flow sensitivity and exploit more outside funds, including both bank loans and trade credit. These effects are more pronounced among financially constrained firms. Even controlling for connections gained through financial institutions or government offices, the effect of diversity still remains very strong. Finally, we conduct several tests to mitigate the concern that our results are driven by the endogeneity of CEOs' appointments.

Key words: CEO, Career Experiences, Corporate Investment, Financial Constraint, social connections

JEL Classification: G31 G32 G39

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1. Introduction

Recently, financial economists have started to acknowledge the influence of manager-specific attributes on firm behaviors. Bertrand and Schoar (2003) document that a manager's fixed effects are important determinants of a wide range of corporate decisions. Inspired by their work, the following studies investigate how managerial characteristics, typically psychological factors such as overconfidence and personal risk attitudes, shape corporate financial decisions and affect firm value.¹

Practically, studies in management fields (e.g. Hambrick and Mason, 1984) have long recognized the impact of managerial characteristics on organizational outcomes. Hambrick (2007) claims managers' experiences, values, and personalities greatly influence their interpretations of the situations they face and, in turn, affect their choices. Managers carry what they have had during their careers as part of their cognitive and emotional givens. The givens serve to filter and distort the decision maker's perception of a particular situation and how it should be handled. Therefore, career experiences can be expected to have a significant effect on the types of actions taken by a manager. A few recent studies in finance shift the focus towards managers' careers. For instance, industry-specific experiences matter for successful acquisitions (Custódio and Metzger, 2013) and corporate divestiture decisions (Huang, 2014). General managerial skills based on past work experiences is an important determinant of CEO pay (Custódio, Ferreira, and Matos, 2013). However, evidence is still scant on the effect of managers' career experiences.

This paper aims to explore the role of CEOs' career experiences in corporate investment and financing decisions. We argue that diverse career experiences help to accumulate social connections. These social connections can mitigate information asymmetry and lead to better access to outside capital and less dependence on internal funds. Therefore, we propose that the investments by a CEO with diverse career experiences are less likely to be constrained by insufficient internal funds and are thus less sensitive to internal cash flow than investments by CEOs with less diverse career experiences.

There are several plausible ways that social connections can facilitate a firm's access to external financing. First, according to the resource-based view, social connections constitute valuable organizational resources (Granovetter, 1985) because they broaden sources of information and

¹ See Malmendier and Tate (2005, 2008), Malmendier, Tate, and Yan(2011), Cronqvist, Makhija, and Yonker(2012), Graham, Harvey and Puri (2013),for example.

improve information quality, relevance, and timeliness (Adler and Kwon, 2002). Diversely experienced CEOs tend to expand a firm's accessible resources by exploiting personal social connections. Second, social connections benefit firms that seek financing from banks through the transfer of private information and promotion of distinguished governance mechanisms (Uzzi, 1999). This would encourage diversely experienced CEOs to utilize external financing when making investment decisions. In addition, the reputation of diversely experienced CEOs among bankers, customers, and suppliers will help firms obtain or retain business relationships and provide indirect financial support, especially when they experience a shortage of working capital. Third, in an emerging market where formal institutions such as laws and regulations are weak, managers and firms have to perform basic functions by themselves. These functions might include obtaining market information, organizing resources, and enforcing contracts (Peng and Luo, 2000). Therefore, we expect that interpersonal connections play an important role in obtaining external financing resources in emerging markets.

We empirically investigate the effect of CEOs' career experiences on corporate financial decisions using a sample of Chinese firms. Our sample includes 563 firms and 1,332 CEOs from 2000 to 2010. We hand-collected biographical data for the CEOs from corporate yearbooks, extracting detailed information on all work organizations where the CEO was employed or served before he/she rose to his/her current position. Specifically, we construct two layers of measures to explore the effect of CEOs' career experiences. In order to reveal the overall diversity of one's career experiences, the first layer of variables measures the number of work organizations (*danwei* in China) for which the CEO has worked. Our sample confirms that overall diversity is an important dimension for capturing the cross-sectional differences of CEOs' career experiences. The second layer of variables stratifies the abstract diversity measure into specific career experiences by constructing dummy variables to indicate whether a CEO has cross-industry experience, or experience in government, financial institutions, or research institutes. These variables are motivated to manifest the heterogeneous effects of career experiences.

Beginning with Fazzari, Hubbard, and Petersen (1988), finance literature attributes the existence of investment-cash flow sensitivity (controlling for investment opportunities) to imperfections in the capital market or information asymmetry between corporate insiders and capital markets. That is,

expensive external financing due to financing frictions may prevent firms from investing in good projects due to insufficient internal funds. In order to investigate whether CEOs' diverse career experiences reduce the frictions, we add the overall diversity measure and its interaction with cash flow to the classic investment equation. The results show that the coefficient of the interaction term is negatively significant, indicating that the investments of CEOs who have more diverse career experiences are less sensitive to internal cash flow. The negative relationship between the diversity of CEOs' career experiences and investment-cash flow sensitivity remains significant even after controlling for investment opportunities, corporate governance, other observable CEO characteristics, industry characteristics, and firm fixed effects. Furthermore, we find connections gained through experiences in government, financial institutions, research institutions or different industries also reduce firms' dependence on internal funds. Interestingly, the effect of political connections or finance background is substantially weakened when controlling for the overall diversity measures, whereas the effect of diversity remains very strong. This finding suggests that overall diversity is a broad measure for capturing connections embodied in various career experiences.

Further, we investigate whether CEOs' career experiences matter more when firms are facing financial constraints. By separating firms according to proxies for financial constraints (including the KZ index, a firm's tangibility, and the ratio of credit to loans), we find supporting evidence that the negative relationship between the diversity of CEOs' career experiences and the sensitivity of investment to cash flow is stronger among financially constrained firms. This finding indicates that connections embodied in CEOs' diverse career experiences indeed mitigate a firm's information asymmetry and financing frictions. In additional analyses, we find that both bank loans and trade credit are positively related to our measures for CEOs' career experiences and that these effects are more statistically significant for financially constrained firms. This evidence suggests that the CEOs with more diverse career experiences are more likely to exploit external debt financing.

A key concern for any analysis of CEO effects is the endogeneity of CEO appointments. In particular, causality may be the reverse, and firms' financing conditions may determine their choice of CEO. Our detailed data allows us to better address these concerns in several ways.

First, we identify the effect of CEOs' career experiences on corporate investment and financing decisions from CEO turnover events. Compared to purely cross-sectional studies, this strategy tends

to be better at distinguishing managerial influences from firm-invariant characteristics, as it infers managerial value from differences in corporate behaviors within the same firm. A similar strategy has been widely used in recent studies on CEO influences and corporate policies (e.g. Malmendier, Tate, and Yan, 2011; Hirshleifer, Low, and Teoh, 2012). Moreover, the 11-year time series provides sufficient variation in CEOs' career experiences to identify CEO effects even after controlling for firm fixed effects and their interactions with cash flow. Thus, the estimated effect does not reflect time-invariant firm characteristics.

Second, CEO turnover may not be a random event, and thus the results may be affected by time-varying firm characteristics. To mitigate this concern, we use the Heckman (1979) two-step procedure to examine whether the factors that lead a firm to choose a diversely experienced CEO in the first instance also drive the lower investment-cash flow sensitivity. We add the likelihood that a firm hires a diversely experienced CEO at a given time into the basic regression to directly control for the endogeneity of CEO selection. However, after correcting for the selection bias, the conclusion is qualitatively the same as before: the diversity of CEOs' career experiences is negatively related to investment-cash flow sensitivity but positively related to the level of firms' external debt financing.

Third, this paper further mitigates the concern of endogeneity by exploiting exogenous shocks on firms' bank financing conditions (the change of total credit regulated by the central bank of China). We find that firms with CEOs that have diverse career experiences are less affected by shocks from macro credit policy. Through this analysis, we not only control for time-invariant firm heterogeneity through adding firm fixed effects and their interactions with changes of macro credit policy in the regression, but also rule out the possibility of firms' time-varying strategic selection of the CEO by exploiting macro credit shocks that are beyond the control of firms' decisions.

Finally, we also attempt to exploit the instrumental variable approach to address the endogeneity concern of CEOs' appointment. That is, we find variables that affect the type of CEO a firm hires but have no effect on corporate financial decisions. In China, a city can be classified into "vice provincial cities" and "ordinary cities," according to bureaucratic rank. Vice provincial cities have competitive advantages in attracting talent because those cities not only enjoy better health, education, and other public services than ordinary cities, but also offer hidden benefits for the personal career development by virtue of their favorable political status. Therefore, it is easier for

firms located in vice provincial cities to recruit diversely experienced CEOs than those in other cities due to the preferred bureaucratic rank. In section 6.3, we argue that after controlling for the city-level economic development, corporate headquarters location (whether it is in the vice provincial city) could be a valid instrumental variable for the type of CEO a firm hires. Then we re-estimate the impact of the diversity of career experiences on corporate investment and financing decisions using the instrumental approach. Our key findings still hold.

Classical corporate finance theory illuminates that the sensitivity of corporate investment to cash flow results from capital market imperfection or information asymmetry (e.g. Hubbard, 1998). In this paper, we find that this effect is mitigated by CEOs' personal characteristics reflected in their career experiences. From this perspective, our paper contributes to the literature that underscores the impact of managerial characteristics on corporate financial decisions. Most prior research in this strand of literature focuses on how a CEO's psychological factors (e.g. overconfidence, risk attitudes) influence corporate decisions and firm value.² A few of them have begun to emphasize the role of CEOs' career experiences. For example, Huang (2014) documents that CEOs' industry experiences have a first-order effect on corporate refocus decisions and that a better match between managerial industry expertise and firms' assets improve firm value. This paper shows that CEOs with more diverse career experiences are more likely to exploit external financing and thus exhibit lower investment cash flow sensitivity. In particular, our measure is similar to the study by Custódio, Ferreira, and Matos (2013) and also exploits mobility across firms and industries (as well as experiences in non-business sectors). Whereas they focus on the general ability associated with CEOs' diversified professional careers, we underscore social connections accumulated through mobility across work organizations. In a broad sense, our evidence that CEOs with diverse career experiences could mitigate information asymmetry and reduce financial constraints complements a specific reason as to why generalist CEOs have higher pay than specialist CEOs.

Our paper is also related to the literature that explores the economic implications of social connections on firms' outcomes. For instance, Faccio, McConnell, and Masulis (2006) document that

² For example, Malmendier and Tate (2005, 2008) find that overconfident CEOs have a heightened sensitivity of investment to cash flow and are more likely to undertake value-destroying mergers. Graham, Harvey and Puri (2013) provide additional evidence that corporate decisions are related to a CEO's personal traits such as risk aversion and optimism, as measured by psychometric tests. Other managerial characteristics that have been documented to affect corporate financial decisions include CEOs' early life experiences, such as military experience and depression-era upbringing (Malmendier, Tate, and Yan, 2011), and a CEO's personal leverage in recent primary home purchases (Cronqvist, Makhija and Yonker, 2012).

politically connected firms have a higher likelihood of receiving bailout assistance. Engelberg, Gao, and Parson (2012) document that employees' personal connections with banks help firms receive more favorable financing terms and thus reduce the cost of debt capital. Consistent with those studies, we find experiences in government offices or financial institutions indeed matter for corporate financial decisions. However, when controlling for the overall diversity measure, the effect of political connections or finance background is substantially weakened while the impact of diverse career experiences remains very strong. This finding not only suggests that the role of diverse career experiences is independent of the effect of political/financial connections that previous literature emphasizes, but raises the possibility that the effect of political/financial connections stems from the overall social connections gained through diverse career experiences rather than rent-seeking activities. Moreover, we focus on the overall connectedness of a specific person (CEO) rather than the social connections between firms. Previous studies do not distinguish the CEO-level choices from firm-level decisions. CEOs' decisions greatly influenced by their career experiences are not necessarily consistent with the optimal choice of the firm as a whole.

The rest of the paper is organized as follows. Section 2 describes the construction of our key variables and research design. Section 3 introduces our sample and reports summary statistics. Section 4 examines the relation between CEOs' career experiences and investment-cash flow sensitivity. Section 5 provides evidence on outside debt financing. Section 6 addresses the endogeneity problems of CEOs' appointment in several ways. Section 7 concludes.

2. Empirical design

2.1 Institutional background and measures for CEOs' career experiences

One's career experiences include three different types of work organizations. These three types are "production," such as a firm; "non-production," such as a university, and "administrative or government organizations." In China, all these work organizations are referred to as "*danwei*."

This paper explains the effect of CEOs' career experiences from the perspective of social connections. Mounting sociology and management theory literature elaborate on the link between career experiences and social connections. CEOs who have worked in different firms bring with them not only knowledge gained through personal experiences with other firms' policies and

practices, but relationships with former contacts and associates (Granovetter, 1985; Geletkanycz and Boyd, 2011) and tend to retain external communication links (e.g., Virany, Tushman, and Romanelli, 1992). Moreover, one's diverse career experiences indicate an extensive circle of weak ties. Individuals with broad weak ties are expected to have an advantageous position in obtaining the right information and achieving good performance (Granovetter, 1973). Additionally, using career experiences information to construct empirical measures for social connections has found favor in strategy and organization studies (Nahapiet and Ghoshal, 1998; Hillman and Dalziel, 2003; Haynes and Hillman, 2010).

Similarly, in China, each *danwei* represents a specific social network that not only refers to the members of the organization but also involves its external connections. For example, the social network of a government office may include members in the office, the *danwei* under its jurisdiction, and the *danwei* that governs it. The social network of a factory may include members of the organization itself, its suppliers, its consumers, and its bankers. In sum, one's career experiences reveal their external networks, and CEOs who have diverse career experiences are likely to possess advantages in external social connections.

Specifically, we construct two layers of measures to explore the effect of CEOs' career experiences. The first layer of variables measures the overall diversity of one's career experiences. The second layer of variables stratifies the abstract diversity into specific career experiences by constructing dummy variables to indicate whether a CEO has certain experiences.

First, we measure the overall diversity of a CEO's career experiences (*#danwei*) as the number of *danwei* the CEO has worked for throughout his or her career as reflected in his or her curriculum vitae. We include both the firm where he or she served and any other concurrent positions held in other firms (e.g. board directors at other corporations) before he or she rose to the CEO position of the focal firm. The number of *danwei* in which the CEO has had experience is calculated in two ways. In the first method (*#danwei*), we count as one *danwei* the different *danwei* in the same business conglomerate or with same administrative functions. In the second method (*#danwei2*), we aggregate the number of *danwei* regardless of whether they were in the same conglomerate or had the same administrative functions. Because the social network of the two *danwei* in the same sector was usually highly overlapped, cross- conglomerate/sector type job mobility should be more valuable

for the accumulation of social connections. Given this, we focus on the measure *#danwei*; however, results are similar when using the measure *#danwei2*. Considering the possible non-linear impact of the number of *danwei* one experienced on the accumulation of social connections, we also define the dummy variable *D_#danwei* (*D_#danwei2*) which equals one if *#danwei* (*#danwei2*) is more than 3 (4) and zero otherwise.³

Second, we decompose the abstract diversity measure into specific career experiences in order to further investigate what kind of career experience matters for corporate investment. Not all work organizations (*danwei*) are identical. The impacts of experiences at each *danwei* are heterogeneous and contingent on its category, sector, and other unobservable factors. While it is difficult to quantify the heterogeneity of each *danwei*, some of them are of particular significance, especially for the accumulation of social connections. For example, those who used to work in banks or other financial institutions may have an advantage in obtaining external financing due to their connections with these institutions. Compared to counterparts in other countries, the Chinese government exerts a powerful influence on many aspects of the economy and society (Fan, Wong, and Zhang, 2007; Li et al., 2008; Chen et al., 2011). Those who worked in government bring with them relationships with government officers and tend to retain links with key people even after they leave the work organization. Moreover, due to the special role of the government in China, career experiences in government offer one the opportunity to have contact with people from all walks of life. CEOs' career experiences in research institutions can act as a kind of endorsement when firms invest in projects that involve a lot uncertainty and need financing from banks or other financial institutions, because those with research experience are considered prestigious and social elites due to their knowledge and professionalism. CEOs that have cross-industry experiences are expected to possess social connections with other industries and thus contribute to expanding the existing network of firms. Besides, possessing such specific experiences indicates a higher probability of having more diverse career experiences. Finally, we construct indicator variables to capture CEOs' social connections obtained from government, research institutions, other industries, financial institutions, and overseas experiences. *D_Gov* (*D_NBS*, *D_#industry*, *D_Fin*), equals one if the CEO had positions in government (academic institutions, more than one Shenwan Level I industry, financial

³ Different thresholds give similar results.

institutions) and zero otherwise (See Appendix for detailed definitions).

One might argue that these measures of diversity signal a CEO's willingness to bear risk given that switching employers usually involves a lot of uncertainty. For example, CEOs with diversified career experiences may be a less risk-averse person and be more willing to increase the leverage of the firm and exhibit lower investment-cash flow sensitivity. Generally, we admit the effect of CEOs' career experiences could be related to cognitive or psychology factors. However, using a Chinese CEO sample can help us mitigate this concern because in a planned economy, a CEO's career path is not entirely the result of self-selection; rather, it is considerably determined by administrative arrangements. Without personal willingness to determine one's career path, one's diversified experiences would be less associated with risk-taking preference. Therefore, we prefer to focus on the perspective of social connections rather than personal psychological factors to account for the effect of diversified career experiences.

2.2 Financial constraint measures

Testing our hypotheses requires separating firms according to an a priori measure of the financial constraint they face. The literature has developed several indices to proxy for financial constraints. Some examples are the KZ index developed by Kaplan and Zingales (1997), a modified version of the KZ index (Baker et al., 2003), and the WW index constructed by Whited and Wu (2006).

Our first proxy for financial constraints is the modified KZ index. Following Baker et al. (2003), we construct the four variable KZ index for each firmyear as the following linear combination:

$$KZ\ index = -1.002 * Cash\ flow - 39.368 * Dividends - 1.315 * Cash + 3.139 * Leverage$$

Where *Cash flow* is earnings before interest, tax, depreciation, and amortization (EBITDA); *Dividends* refers the cash profits that firms distribute to shareholders in a given year; *Cash* is a firm's year-end cash holdings; all variables above are scaled by beginning-of-year total asset; and *Leverage* is the ratio of a firm's total liabilities to beginning-of-year total assets.

However, the applicability of these indices is controversial. Hadlock and Pierce (2010) find that the components of these indices are, indeed, endogenous financial choices that may not have a straightforward relation to financial constraints.⁴ In China, bank loans are the primary source of

⁴ For example, while an exogenous increase in cash on hand may help alleviate the constraints that a given firm faces,

external finance for firms. A firm that has a higher likelihood of obtaining bank loans at a reasonable cost is less likely to be confronted with financial constraints. We thus choose other variables that are directly associated with financial status (especially the likelihood of obtaining bank loans) to proxy for firm-level financial constraints.

The second variable is asset tangibility. As argued by Almeida and Campello (2007), firms with greater tangibility are less likely to be financially constrained, because tangible assets mitigate contractibility problems; that is, tangibility increases the value that can be captured by a creditor in default states. Moreover, in China, using tangibility as the criteria for the classification of financial-constraint status has intuitive appeal because it represents a direct measure of the scale of accessible bank loans and the premium paid by firms.⁵ Following Almeida and Campello (2007), Tangibility is calculated as follows:

$$\text{Tangibility} = (\text{Cash} + 0.715 * \text{Receivables} + 0.547 * \text{Inventory} + 0.535 * \text{Capital}) / \text{Total Assets}.$$

Here *Cash* is a firm's year-end cash holdings, *Receivables* includes the firm's accounts receivable and other receivables, *Inventory* is the firm's year-end inventory, *Capital* is the value of net property, plants, and equipment, and *Total Assets* is a firm's book value of total assets. After the value of *Tangibility* is calculated, a firm whose tangibility is less than the sample median is classified as financially constrained and as financially unconstrained otherwise.

The third variable is the credit loan ratio, defined as the percentage of credit loans in the total amount of loans. Theoretical research suggests that lines of credit are critical to reducing future capital market friction facing firms (Almeida et al., 2004). Sufi (2009) provides evidence that lack of access to a line of credit is a more statistically powerful measure of financial constraints than traditional measures used in the literature. In this sense, firms that are qualified to obtain a substantial proportion of credit loans are less likely to suffer from financial constraints. We define firms as “unconstrained” when their credit loan ratio is above that of the median firm. Firms that do not meet this criteria are designated “constrained.” To account for the changing levels of financial constraints over time at the firm level, we allow reclassification of a firm's financial status every year, and group

the fact that a firm chooses to hold a high level of cash may be an indication that the firm is constrained and is holding cash for precautionary reasons. Since the China Securities Regulatory Commission (CSRC) examines and approves firms' CEOs by taking into account their payout records, firms paying dividends are likely to refinance through capital markets subsequently. Therefore, in China, paying dividends is not a reliable signal concerning the firm's financial status.

⁵ In China, most banks follow mortgage-based covenants to provide loans.

composition is permitted to vary every year.

Since the variables above are not choice variables for managers in the short run and are unlikely to depend on investment over the short time period covered by our panel, we can regard them as exogenous. Even though it is doubtful that using either of these variables to sort firms into putatively constrained/unconstrained groups will misclassify some firms, we use the three variables simultaneously. If the results are robust to different measures of constraint, we can declare that our results are not distorted by our classification procedure.

2.3 Factors affecting investment-cash flow sensitivity

2.3.a Investment Opportunities

Firms' growth opportunities would affect the link between investment and cash flow. It is likely that the investment-cash flow sensitivity may be simply due to Tobin Q's measurement errors concerning marginal investment opportunities (Erickson and Whited, 2000). Therefore, we include Tobin Q and sales growth⁶ and their interactions with cash flow to control for the effect of investment opportunities.

2.3.b Ownership Structure

If an agency problem prevails in our sample, the sensitivity of investment to cash flow varies with the corporate governance factors. Interest conflicts between controlling shareholders and small shareholders have been considered one of the most important corporate governance issues in Asian stock markets (Fan, Wei, and Xu, 2011). A large body of literature documents that ownership structure does affect investment-cash flow sensitivity (e.g. Wei and Zhang, 2008), while the underlying mechanism is somewhat complicated and ambiguous.

The first corporate governance variable we consider is the ownership of the largest shareholder. On the one hand, managerial entrenchment may be reduced because a large shareholder has incentive to perform an active monitoring role (Grossman and Hart, 1980). Hence, we expect that block shareholder monitoring reduces agency costs and investment-cash flow sensitivity. On the other hand, the ability to control shareholders to expropriate the minority shareholders is directly related to the degree to which they control the company. Greater ownership by controlling shareholder

⁶ Sales growth has been widely used as a measure of investment opportunities to avoid the measurement error. See for example, Whited and Wu (2006).

corresponds to a higher incentive to tunnel (Lemmon and Lins, 2003). In this view, we expect that the ownership of the largest shareholder increases investment-cash flow sensitivity. We include ownership nature (a dummy indicates whether the firm is a state-owned enterprise), the ownership percentage held by the largest shareholder, and their interactions with cash flow in the regressions.

Moreover, the ability of a controlling shareholder to expropriate is also related to the challenges they face from minority shareholders. Accordingly, we define the second corporate governance variable as a Herfindahl-type index that measures the concentration of shares held by the top ten shareholders after excluding the largest shareholder. In firms with dispersed ownership structure, entrenchment behaviors are more likely to occur due to a lack of effective monitoring. Hence, we expect that a dispersed ownership structure would strengthen investment-cash flow sensitivity.⁷

2.3.c Other CEO Characteristics

We also control the relation between the diversity of CEOs' career experiences and other observable executive characteristics: educational background, tenure, age, and so on. We analyze their effects on investment-cash flow sensitivity and ask whether the diversity of CEOs' career experiences affects investment decisions independently.

Malmendier and Tate (2005) document that overconfident managers display higher investment-cash flow sensitivity because they overestimate the returns on their investment projects and are reluctant to raise external funds when they do not have sufficient internal funds. The overconfidence hypothesis could easily be extended to incorporate other managers' personal characteristics. For example, managers with an MBA degree are more likely to be overconfident, possibly because an MBA is perceived as the best general management degree (Bertrand and Schoar, 2003; Beber and Fabbri, 2011). Therefore, we expect that the investment of a CEO with an MBA degree is more responsive to cash flow.

Age and tenure can also affect the sensitivity of investment to cash flow for reasons related to risk attitudes because risk-averse CEOs are less likely to level up the firm when there are insufficient internal funds. For example, older executives have greater costs of failure because getting re-employed is more difficult. Similarly, executives that have been employed by the firm for a longer

⁷ The results do not change when we include other corporate governance variables such as the ratio of independent directors and managerial ownership. To save space, we do not report these results, though they are available on request.

time have less need to establish a reputation and therefore become more risk averse (e.g. Gibbons and Murphy, 1992; Graham et al., 2013). In this view, younger managers with fewer years' work experience are expected to display less investment-cash flow sensitivity.⁸

The last control variable is outside/inside hiring of a CEO, i.e. whether the CEO is rising through the ranks within the firm or is hired from the outside. By definition, there is some correlation between the measures for diversity of CEO career experiences and the outside succession dummy. Therefore, we can expect that a CEO hired from outside could reduce the firms' investment-cash flow sensitivity due to his/her advantage in external financing.

2.4 Empirical specifications

To test whether the sensitivity of investment to cash flow decreases with the diversity of CEOs' career experiences, we use the following general regression specification:

$$I_{it} = \beta_1 + \beta_2 CF_{it} + \beta_3 Q_{it-1} + \beta_4 V_{it} + \beta_5 V_{it} * CF_{it} + \beta_6 CF_{it} * Q_{it-1} + X'_{it} B_7 + CF_{it} * X'_{it} B_8 + \varepsilon_{it} \quad (1)$$

where the dependent variable I is a firm's investment expenditure in a year, which is measured as cash payments for fixed assets, intangible assets, and other long-term assets from the cash flow statement, minus cash receipts from selling these assets, scaled by beginning-of-year total assets. CF is cash flow that is measured as a firm's current period net operating cash flow, scaled by total assets the beginning of the year. Q is a firm's *Tobin Q* at the beginning of the year. The 'q-theory' of investment implies that incremental investment is an increasing function of the marginal Tobin Q . In general, firms with more valuable growth opportunities are likely to invest more. As in previous studies on China (Chen et al., 2011), we define Tobin Q as the sum of the beginning market value of owner equity and the book value of total liabilities, divided by the beginning book value of total assets.⁹

The main variable of interest here is V , i.e. measures for CEOs' career experiences. The coefficient of the interaction term $V*CF$ in Equation (1) reflects the effect of a CEO's career experiences on a firm's investment-cash flow sensitivity. X here refers to investment opportunities, corporate governance factors, and other CEO personal characteristics mentioned in a previous

⁸ However, we do not find significant influence from age or tenure. To avoid the estimation biases from co-linearity, we do not include them in the regression.

⁹ Following Villalonga and Amit (2006), we multiply the total shares outstanding (including tradable and non-tradable shares) by the share price of tradable shares to estimate the market value of common equity to measure Tobin Q alternatively. The results (unreported) are qualitatively unchanged.

section. In this paper, we focus on the sensitivity of investment to cash flow rather than the level of investment. Therefore, their interactions with cash flow are also included in the equation. We include the inverse of total assets to isolate the correlation between a dependent variable and independent variable induced by a common scaling variable. To control for industry (firm) fixed effects, we add an industry (firm) dummy and its interaction with cash flow in the regression. To reduce the influence of outliers, each of these continuous variables is winsorised at the 1st and 99th percentiles. The standard errors are computed by clustering the observations within each firm. This process treats the time series of observations within the firm as a single observation, effectively eliminating any serial correlation.

We further partition our sample based on the degree of financial constraint to investigate whether the impact of CEOs' career experiences on investment-cash flow sensitivity varies across firms with different financial conditions.

3. Sample and data

To investigate the effect of a CEOs' career experiences on corporate investment, we select all CEOs of firms on the Zhong-Zheng 800 index (a popular index in China's A share market) at the end of 2010 as an initial sample and exclude the CEOs of financial firms and the firms that became public after 2004.

We hand-collected biographical data for the CEOs from yearbooks, extracting detailed information on all firms where the CEO was employed or served on the board of directors before he/she rose to the current position. When CEOs' biographical information was missing in the yearbooks and the Sina finance website (<http://finance.sina.com.cn/stock/>), we deleted the firm. The final sample includes 563 firms and 1332 CEOs from 2000 to 2010.

Table I reports the descriptive statistics of the variables used in our empirical analyses. Panel A presents the summary statistics of CEOs' career experiences. In our sample, a CEO, on average, worked for two different employers before he or she rose to the current position. If different *danwei* under the same business conglomerate or sector are counted as different *danwei*, this number grows to four. Thirty-eight percent of the CEOs in our sample have cross-industry work experience, 17% of them worked in government, and 16% of them worked in academic institutions. However, the

percentage of CEOs who worked in financial institutions is no more than 8%. Additionally, one third of CEOs in our sample were directly hired from the outside.

Panel B provides frequency statistics of key measures (*#danwei* and *#danwei2*) for the diversity of CEOs' career experiences. Taking the frequency of *#danwei* for example, it is a bit surprising that 33% of CEOs have stayed with one firm since the start of his or her career. While half of CEOs have worked for no more than two *danwei*, 30% of CEOs have worked for four or even more *danwei*. This fact suggests the diversity of career experiences is an important dimension for capturing the cross-sectional differences of CEO personal characteristics.

Panel C describes the Spearman correlations between CEO characteristics. *D_#danwei* (*D2_#danwei*) is an indicator variable that is one if *#danwei* is greater than three (two) and zero otherwise. The correlations between *D_#danwei* and the dummy variables that indicate cross-industry or government experiences are relatively high and above 0.35, while the correlations between *D_#danwei* and other typical experiences are also significantly positive with the coefficients around 0.1. This evidence suggests that CEOs who had cross-industry experiences or worked in government or academic institutions are likely to have more diverse career experiences. As expected, there is positive correlation between *D_#danwei* and *Outside* but the magnitude is relatively moderate at about 0.15.

We merge the CEO data with firm-level financial and accounting variables from 2000 to 2010. Accounting and financial data are obtained from the China Stock Market & Accounting Research (CSMAR) database developed by GTA Information Technology, one of the major providers of Chinese data. In panel E, we report the summary statistics of the firm-year level variables used in our analysis. The Appendix provides detailed definitions of all the variables listed. All continuous variables are winsorised at the 1st and 99th percentiles.

Insert Table I here

We also split the sample into firms with and without diversely experienced CEOs. Table II reports the mean and median of corporate characteristics, respectively, for firms with diversely experienced CEOs and those with simply experienced CEOs. Simply experienced CEOs are defined as those who worked for no more than three employers before joining the current firm, while diversely experienced CEOs refer to those who worked for no less than four firms. The significance

of the mean/median difference between these two groups of firms is also reported. Basically, we find that the investment and cash flow are not significantly different (in terms of median) among these two subsamples, nor are ROA or total assets. Thus, the summary statistics fail to reveal a systematic pattern of higher profitability or larger firm size for firms with diversely experienced CEOs, to some extent revealing a random sorting of diversely experienced CEOs. In addition, Tobin Q, sales growth rate and leverage are higher for firms with diversely experienced CEOs, though all the differences are relatively small.

Insert Table II here

4. CEOs' career experiences and investment-cash flow sensitivity

Table III reports basic regression results to demonstrate the effect of CEOs' career experiences on investment-cash flow sensitivity. The dependent variable is investment scaled by lagged period total assets. The independent variables only include Cash flow, lagged Tobin Q, and career experience measures and their interactions with Cash flow.

Panel A displays the results using the overall diversity measures of CEOs' career experiences. In column (1), the measure is based on the number of *danwei* for which the CEO has worked and where different firms in the same sector are counted as one firm, while the measure in column (2) is similar to the first column with the exception that different *danwei* in the same sector are counted separately. Career experiences measures in column (3) and column (4) are dummy variables based on the number of *danwei* calculated in the first method with 3 and 4 as the threshold, respectively. Industry fixed effects and their interaction items with cash flow are included in the first four regressions. First, our results confirm the stylized facts of the investment-cash flow sensitivity literature, namely, that cash flow has great explanatory power beyond Q for investment. Investment opportunities measured by Tobin Q also have a significant and positive impact on investment, while the coefficients of their interactions with cash flow are not significant. Consistent with the prediction when interpreting the overall diversity measures from the perspective of social connection, the key result is that the interaction of overall diversity measures and cash flow are all significantly negative, indicating that the investments of CEOs who have more diverse career experiences are less sensitive to internal cash flow. This result suggests the characteristics of CEOs'

career experiences matter for corporate investment. We also find that CEOs with more diverse experiences tend to invest less (although this is only significant in some cases), indicating that diversely experienced CEOs are reluctant to invest inside the company.

Columns (5) to (8) present the basic results after including firm fixed effects and their interaction items with cash flow in the regressions. This specification identifies the effect only from time-series variation within the firm. Fortunately, our data provides sufficient variation in CEOs' career experiences to identify CEO effects even after controlling for firm fixed effects and their interactions with cash flow. For example, consider the measure (D_danwei). In 163 cases of our sample, the D_danwei dummy variable changes from zero to one, and in 198 cases from one to zero. The value of D_danwei shows time series variation in 323 out of 563 firms. Thus, the estimated effect does not reflect time-invariant firm characteristics. The results show that the coefficients of the interaction items between our diversity measures and cash flow are still significantly negative. The robustness to the inclusion of the firm fixed effects and their interactions with cash flow eliminates any alternative explanation for our results that relies on fixed cross-sectional differences across firms with and without diversely-experienced CEOs. Given that the results using various diversity measures are similar, we only report the one using D_danwei as the proxy for the overall diversity in the following tables.

Panel B investigates the impacts of specific career experiences on corporate investment by decomposing the overall diversity measure. Career experience measures in Column (1) through Column (4) are indicator measures for whether the CEO has a cross-industry or government background, finance background, or research experience, respectively. We include firm fixed effects and their interactions with cash flow in all regressions. The results show that CEOs with work experiences in government, financial or research institutions, or cross-industry exhibit lower investment cash flow sensitivity than those without such experiences. Similar to the effect of diverse career experiences, we find social connections gained through experiences in government, financial or research institutions, or cross-industry can reduce firms' dependence on internal funds.

Furthermore, we compare the effect of the diversity measures with that of each specific experience because results in Columns (1) through (4) are not clear on the relative importance of specific connections such as political or bank connections and connections embodied in the overall

diversity measure. In Columns (4) through (8), we therefore simultaneously include a diversity measure, dummy variables of specific experiences, and their interactions with cash flow in the regressions.

Interestingly, we find that the effect of political (financial) connections is substantially weakened and turns insignificant when our overall diversity measure (D_danwei) is added to the regression, whereas the impact of diverse career experiences is still very strong. This result, on one hand, shows that the role of diverse career experiences is independent of the effect of political (financial) connections that previous literature emphasizes; on the other hand, it raises the possibility that the effect of political (financial) connections on corporate investment stems from the accumulation of overall social connections through the mobility across work organizations rather than rent-seeking activities with government or bank officers. At the very least, although a CEO's political (financial) connections may be important for determining investment policies, their effects could be dominated by CEOs' overall connections.

Insert Table III here

Table IV reports the results after we include all the control variables. The results demonstrate that connections embodied in CEOs' career experiences can still significantly reduce investment-cash flow sensitivity after controlling for the impact from investment opportunities, corporate governance, and other observable characteristics of CEOs. The effect of Tobin Q interacted with cash flow is significantly negative, consistent with agency-related interpretations. However, we find a significant positive relationship between sales growth and investment-cash flow sensitivity, which instead can be better understood in the framework of information asymmetry. Though this result is difficult to interpret, it is not relevant to our main focus. Private firms exhibit lower investment-cash flow sensitivity than state-owned firms, indicating that the agency cost induced by free cash flow is more severe among state-owned firms. To avoid the multicollinearity problem, we include one corporate governance variable each time. Firms with higher ownership by controlling shareholders tend to invest more, while the ownership of the controlling shareholder has no impact on investment-cash flow sensitivity. Firms with a more dispersed ownership structure not only tend to invest more but also display higher investment-cash flow sensitivity. One possible interpretation of this result is that a dispersed ownership structure has insufficient incentive to

discipline the CEO's power and thus results in more severe empire-building activities. To save space, we only report the regression results including *D_Block* and its interaction with cash flow to control for the impact of corporate governance. The results do not qualitatively change when substituting them with other governance factors such as *D_Shr2_10* and *D_CEOshr*. As for CEOs' other characteristics, CEOs with an MBA degree exhibit higher investment-cash flow sensitivity, consistent with the overconfidence hypothesis.

Insert Table IV here

Table V reports the effect of financial constraints on the relationship between investment-cash flow sensitivity and the diversity of CEOs' career experiences. We separate our sample into two groups based on the value of each of three *priori* proxies for financial constraints documented in the previous section. Firms are classified as financially constrained (FC), if their KZ index in the lagged period or tangibility or credit loan ratio is lower than the median of the sample. Otherwise, we deem that they are less likely to be financially constrained (UFC). To account for the change of financial constraints over time at the firm level, we reclassify a firm's financial status every year. Then we run the OLS regression including all the control variables in Table IV to estimate the coefficients of the interaction items of the diversity measure and cash flow.¹⁰

Then, we investigate whether the differences between the two groups are statistically significant. Following Cleary (1999), we use simulation evidence to determine the significance of observed differences in coefficient estimates. A bootstrapping procedure is used to calculate empirical p-values that estimate the likelihood of obtaining the observed differences in coefficient estimates if the true coefficients are, in fact, equal. Observations are pooled from the two groups whose coefficient estimates are to be compared. Using n_1 and n_2 to denote the number of annual observations available from each group, we end up with a total of $n_1 + n_2$ observations every year. Each simulation randomly selects n_1 and n_2 observations each year from the pooled distribution and assigns them to group 1 and group 2, respectively. Coefficient estimates are then determined for each group using these observations, and this procedure is repeated 5000 times. The empirical p-value is

¹⁰ Because the partition greatly reduces the time series variation within each firm, there are an insufficient number of cases in which diversely experienced CEOs and simply experienced ones are hired in the same firm to draw a robust inference from any estimation. The lack of identifiable cases points to a potentially severe sample selection bias from identifying solely out of somewhat anomalous firms with multiple short-tenured CEOs. To avoid such biases, all regressions in Table V include industry fixed effects and their interactions with cash flow.

the percentage of simulations where the difference between coefficient estimates is below the actual observed difference in coefficient estimates. For brevity, we only report the values of β_2 to β_5 . We find, as predicted, that CEOs' career experiences have a greater impact on the sensitivity of investment to cash flow within financially constrained firms. Moreover, in most cases, we can find a significant negative relationship between the diversity of CEOs' career experiences and investment-cash flow sensitivity *only* among financially constrained firms, indicating that career experience-based social connections only matter when firms have insufficient internal funds. In addition, investment-cash flow sensitivity is greater among financially constrained firms. This finding suggests that the investment-cash flow sensitivity is at least partly due to financial constraint.

Insert Table V here

5. CEOs' Career Experiences and Firms' Outside Debt Financing

Previous analyses show that social connections embodied in the diversity of career experiences do alleviate financial constraints. If it is because of the advantage in external financing, we should observe that diversely experienced CEOs are more likely to utilize outside debt financing than their counterparts.

On one hand, in China, obtaining bank loans is the most important formal financing channel even for listed firms (Allen et al. 2005). Uzzi (1999) documents that social relations and networks can benefit firms that are seeking financing from banks through the transfer of private information and promotion of distinctive governance mechanisms. If the career experience-based social connections are valuable, diversely-experienced CEOs should be able to expand the pool of outside financing by virtue of bank loans.

On the other hand, CEOs' career experiences reflect not only their accumulated relationships with bankers or other formal finance providers but also their connections with suppliers and consumers. We argue that the CEO's reputation and relationship among customers and suppliers will help firms obtain or retain business relationships and provide indirect financial support when they experience a shortage of working capital. Trade credit can be the most direct way to achieve this purpose. In fact, it is one of the informal financing mechanisms suggested by Allen, Qian, and Qian (2005) and has become more relevant when considering institutional reasons of the Chinese financial

system's misallocation of formal credit (Cull, Xu, and Zhu, 2009). Besides, trade credit is also the most important form of short-term external finance (even in the United States) due to information advantages that suppliers have over financial institutions in providing credit to their own customers (Peterson and Rajan, 1997). Therefore, we expect that firms with diversely experienced CEOs are more likely to use trade credit to satisfy their financing needs than their counterparts.

Therefore, our study uses both bank loans and trade credit as the dependent variables. Bank loans are defined as the ratio of the proceeds from bank loans to the book value of total assets, while trade credit is the sum of accounts payable and other payables normalized by total assets. In order to isolate the effect of CEOs' career experience on firms' outside financing, this paper follows the capital structure literature and includes firm size, profitability, growth opportunities, asset tangibility and CEO incentive factors as control variables in the regressions.¹¹ We controlled firm and year fixed effects in all regressions.

Table VI reports the impact of CEOs' career experiences on firms' outside financing. Panel A displays the results using bank loans as the dependent variable, while Panel B shows the results using trade credit as the dependent variable. We find that all the estimated coefficients on the diversity measure are positive for both bank loans and trade credit. CEOs that have financial connections display similar patterns. The coefficient of political connections is significant only for bank loans, while the effect of experience in research institutions is significant only for trade credit. These estimates are robust to clustering the observations by firm. There is a positive relationship between the diversity of CEOs' career experiences and firms' outside financing, indicating that diversely experienced CEOs are more likely to use both formal finance (e.g. bank loans) and informal finance (e.g. trade credit). However, the impacts of specific experiences on firms' financing patterns are heterogeneous.

Furthermore, we investigate whether the positive relationship varies depending on the dimension of financial constraint. Similar to the previous methods in Section 4, we separate our sample into two groups based on the value of each of three *priori* proxies for financial constraints documented in the previous section and run the panel regression, including all the control variables

¹¹ Titman and Wessel (1988) and Rajan and Zingales (1995) find that these variables hold for the United States and other developed economies. In addition, Antoniou et al. (2008) find that the aforementioned determinants also work well in developing economies.

in Panel A (B), respectively in each group. Given that there is no significant difference between the results of bank loans and trade credit when using the diversity measure, we use the sum of bank loans and trade credit as the dependent variable. Panel C of Table VI shows that the significant positive relationship between career experience measures and firms' outside financing only exists among financially constrained firms, suggesting that CEOs' career experiences do matter when financially constrained firms seek outside financing.

Insert Table VI here

6. Endogeneity of CEOs' Appointment

A key concern for any analysis of CEO effects is the endogeneity of CEO appointments. In particular, the causality may be reverse, with firms that display low investment-cash flow sensitivity seeking diversely experienced CEOs. The main difficulty in interpreting our findings is unobserved firm heterogeneity. The robustness of our findings to the inclusion of firm fixed effects and their interactions with cash flow rules out any alternative explanation for our results that relies on fixed cross-sectional differences across firms with and without diversely experienced CEOs. However, our results may reflect time-varying firm characteristics. For example, if some unobserved determinants of investment efficiency also change around the top executive turnover, the appointment of a diversely experienced CEO could still be endogenous.

Before we address this problem directly, we note that there is a mismatch between the variation of CEO characteristics and that of investment and financing within firms. It is least likely that the turnover of a CEO is associated with specific investment or financing policy. Furthermore, if a CEO is hired to implement a specific firm policy, its impact should be mainly felt in the first one or two years after the CEO takes office. As a robustness check, we recode the diversity dummy measure as zero in those years. Replicating Table IV and Table VI, we find very similar results.¹²

6.1 The Heckman (1979) two-stage procedure

Following Villalonga and Amit (2006) and Chen et al. (2011), we apply the Heckman (1979) two-stage treatment effect procedure to alleviate the endogeneity concern and reproduce the key results in Table IV and Table VI. This approach mitigates possible biases caused by the

¹² The results are available on request.

correlated-but-omitted-variable problem if such omitted variables change slowly over time.

As a first step, we will investigate which factors affect a firm's CEO choice. Following previous literature, we take into account factors such as prior firm performance (Farrell and Whidbee, 2003; Graham et al., 2013), board structure (Fee, Hadlock, and Pierce, 2011), prior CEO's characteristics (Finkelstein and Hambrick, 1990; Allgood and Farrell, 2003) and industry features (Graham et al., 2013) to predict the likelihood that a firm would hire a diversely experienced CEO. The dependent variable is the diversity measure (D_danwei), and explanatory variables include *Prior CEO tenure*, *Prior forced turnover*, *Prior ROA*, *Prior sales growth*, *Prior independent directors' ratio*, and *Prior block shareholder ownership*, and industry fixed effects. For detailed definitions of these variables, please see the Appendix. One more point worth noting is that here, we use information for the year prior to CEO turnover as instrument variables. A lag between firms' CEO appointment decision and the investment policy of the new CEO alleviates the concern that variables in the selection equation could feasibly play an independent role in investment-cash flow sensitivities.

We find that firms that dismiss their CEOs prefer diversely experienced CEOs to change the policy since a forced turnover suggests a mismatch between the prior CEO and the firm (Allgood and Farrell, 2003). Firms with short-tenured prior CEOs are more likely to hire diversely experienced CEOs perhaps because such CEOs, who had more diverse career experiences and spent less time in each firm on average, possess a managing style similar to that of prior CEOs. Moreover, a larger ownership by controlling shareholders, to some extent, prevents the selection of diversely experienced CEOs. It is somewhat surprising that a higher independent directors' ratio would also reduce the likelihood of hiring a diversely experienced CEO, although the coefficient is insignificant at a reasonable confidence level. Firms' prior profitability and growth rate also have some influence on the new CEO selection, but their directions are not consistent.

We obtain the likelihood that a firm would hire a diversely experienced CEO (Λ) from first-stage probit regression and include it as a control variable in the second-stage analysis. Panel B of Table VII reports the second stage regression results for investigating the relation between the diversity of CEOs' career experiences and investment-cash flow sensitivity. Panel C of Table VII reports the second stage regression results for the leverage equation. To save space, we omit the coefficients of control variables in the table. Mill's ratio, Λ , has an insignificant coefficient,

indicating that selection bias might not be severe in our data. Most important, after correcting for selection bias, the diversity of CEOs' career experiences still negatively related to investment-cash flow sensitivity but positively related to the level of firm external debt financing. Therefore, the conclusion is qualitatively the same as before.

Insert Table VII here

6.2 Exogenous shocks on outside financing

Credit control is one of the major instruments of the monetary policy used by the People's Bank of China ("central bank") to adjust the supply of money (liquidity) in the economy. Central bank administrators have control over the credit granted by commercial banks. Therefore, the degree of difficulty firms have in obtaining financing from banks hinges on the credit controls of the central bank. When the central bank is promoting an expansionary credit policy, it becomes easier for all firms to obtain credit from commercial banks and thus corporate investment is more likely to increase. In contrast, when the central bank implements a credit tightening policy, the investment of firms becomes more sensitive to the availability of internal funds and good projects may be rejected due to limited access to bank financing. Therefore, for a single firm, the change of credit control policy by the central bank can be considered as an exogenous shock to a firm's accessible funds and thus their investment decisions.

We introduce the growth rate of total credit supply (*Credit_gr*) as the proxy for the condition of credit policy.¹³ We expect that firms have to reduce their investment when the central bank implements a credit tightening policy and suppresses the growth rate of the total credit supply. However, if social connections embodied in CEOs' career experiences help the firms to reduce financing frictions, firms with diversely experienced CEOs will be less affected by this policy and display lower sensitivity to macro-policy changes. We add the item *Credit_gr*, and their interaction item with the diversity measure *Credit_gr*D_#danwei* into the determinant equation of investment-cash flow sensitivity to capture the effect of exogenous shocks. Simultaneously, we include firm fixed effects and their interactions with cash flow into the regression to control for the impact of unobservable firm characteristics.

¹³ The data regarding the growth rate of total credit supply is available on the website of the People's Bank of China (<http://www.pbc.gov.cn/publish/english/963/index.html>).

Table VIII reports the regression results. The significantly positive coefficient of the item *Credit_gr* suggests that the overall corporate investment level is greatly affected by macro credit policy. Next, as predicted, the coefficient of item *Credit_gr* D_#danwei* is significantly negative, indicating the critical role of the diversity of CEOs' career experiences in alleviating financing difficulty. These results further mitigate the concern of endogeneity, not only because we control for time-invariant firm heterogeneity through adding firm fixed effects and their interactions with changes of macro credit policy in the regression, but also because we rule out the possibility of firms' time-varying strategic selection of the CEO by exploiting the macro credit shocks that are beyond the control of the firm's decisions.

Insert Table VIII here

6.3 The instrumental estimation

Finally, we also attempt to exploit the instrumental variable approach to address the endogeneity concern of CEOs' appointments. That is, we find variables that affect the CEOs' appointments but have no effect on corporate financial decisions.

In China, a city can be classified into two types, according to bureaucratic rank. These two types are "vice provincial cities," including capital cities of provinces and other important cities (Shenzhen, Xiamen, Ningbo, for example), and "other ordinary cities," (Suzhou, Wuxi, Changzhou, Dongguan, for example). On one hand, compared to ordinary cities, vice provincial cities, due to administrative reasons, enjoy better health, education, and other public services.¹⁴ Better supporting policies attract more talent to concentrate in vice provincial cities. On the other hand, vice provincial cities could offer hidden benefits for personal career development by virtue of their favorable political status. For example, resource allocation controlled by the government is closely associated with bureaucratic rank and is more often than not tilted towards vice provincial cities. Therefore, it should be easier for firms located in vice provincial cities to recruit diversely experienced CEOs than those in other cities. This has been confirmed by our sample. The proportion of diversely experienced CEOs among firms with headquarters in vice provincial cities is 11% higher than that in ordinary cities. Factors that affect talent mobility are least likely to be related to corporate financial decisions.

¹⁴ According to the China City Statistical Yearbook (2011), in Xi'an, the number of full-time university faculties per ten thousand, the number of hospital beds per ten thousand, and the number of professional physicians per ten thousand, are 73, 57, and 23, respectively; in Suzhou, the corresponding values are 31, 48, and 20. However, the per capita GDP of Suzhou (94,270 Yuan) is more than twice that of Xi'an (42,573 Yuan).

One might argue that the bureaucratic rank of a city is related to its level of economic development, which also affects corporate financial decisions. We address this concern in the following ways. First, the bureaucratic rank of a city is not necessarily positively related to its level of economic development. For example, it is quite common that per capita GDP of ordinary cities in East China (e.g. Suzhou, Changzhou, Wuxi) is much higher than that of vice provincial cities (e.g. Xi'an, Taiyuan, Zhengzhou) in middle or west China. Second, the relation between the level of economic development and financing difficulty is ambiguous. On one hand, developed financial services promoted by strong economic growth increase the chances for firms to obtain external financing. On the other hand, strong financing demand in developed areas triggers fierce competition between firms. Thus, the association between regional economic development and corporate financing is likely to be weakened by these two opposing forces. Finally, we add the city-level per capita GDP into the regressions in Table IV and Table VI to control for the impact of city-level economic development on corporate investment and financing decisions. We believe that after controlling for city-level economic development, corporate headquarters location should be related to factors that affect talent mobility, but be irrelevant to regional economic factors that might affect corporate financial decisions. In this section, we take the corporate headquarter location (whether it is a vice provincial city¹⁵) as the instrumental variable for the type of CEO a firm hires, and estimate the impact of the diversity of career experiences on corporate investment and financing decisions.

Table IX reports regression results with the corporate headquarters location (*Capital*) as an instrumental variable for the diversity of CEOs' career experiences. *Capital* is a dummy variable, which equals one when a firm's headquarters is located in a vice provincial city and zero otherwise. Panel A reports the results of investment-cash flow sensitivity. In addition to all control variables in Table IV, we add into the regression the per capita GDP of the city where a firm is headquartered and its interaction with cash flow. In the first-stage regressions, the dependent variables are the overall diversity measure (D_danwei) and its interaction with cash flow, respectively. The results show that a firm headquartered in a vice provincial city is more likely to hire a CEO with diverse career experiences, and that the coefficient of the interaction item between

¹⁵ Beijing, Shanghai, Tianjin, and Chongqing are also classified as the vice provincial cities.

headquarters location and cash flow is positive. The F statistics that test whether the coefficients of instrumental variables are zero, are 10.625 and 25.321, respectively. The results of the second-stage regression show that the coefficient of the interaction item between the overall diversity measure and cash flow is still significantly negative by exploiting the instrumental estimation. The coefficient of the interaction item between the city-level per capita GDP and cash flow is not significant. This evidence suggests that there is no stable relation between the level of economic development and financial constraints possibly because strong financing demand and fierce competition among firms counteract financing convenience in the developed area.

Panel B reports the results for external debt financing using the instrumental estimation. Similarly to Panel A, we include the city-level per capita GDP beside all control variables in Table VI to remove the influence of regional economic development on corporate external financing. In the first-stage regression, the dependent variable is the overall diversity measure. The coefficient of the headquarters location is positive with F statistics greater than 12. The results of the second-stage regression show that CEOs with more diverse career experiences have a heightened propensity for exploiting external debt financing. Overall, the instrumental estimation indicates that the effect of CEOs' career experiences does not stem from firm-level factors. Additionally, the effect of the city-level per capita GDP on corporate leverage is not significant.

Insert Table IX here

7. Conclusions

This paper investigates the impact of CEOs' career experiences on corporate investment decisions. By hand-collecting the biographical information of 1332 Chinese CEOs, we find that firms with CEOs who have more diverse career experiences exhibit lower investment-cash flow sensitivity and exploit more outside funds including both bank loans and trade credit. Further analyses show that firms that hire CEOs that have experiences in financial institutions or government display similar patterns, however, even controlling for such experiences, the effect of diversity still remains very strong. Our findings are more pronounced among financially constrained firms. Finally, we conduct several tests to mitigate the concern that our results are driven by the endogeneity of CEOs' appointments.

In this paper, we interpret the effect of diversity of CEOs' career experiences from the perspective of social connections. Our results show that connections gained through CEOs' diverse career experiences indeed reduce financial constraints and assist firms in obtaining external debt financing. This finding is beyond the effect of political or bank connections. In a broader sense, our evidence suggests that interpersonal ties play a more important role in facilitating economic exchanges in countries in which formal institutions such as laws and regulations are weak.

Our findings also suggest that the diversity of CEOs' career experiences is an important dimension for capturing CEOs' personal characteristics and could exert systematic impacts on corporate investment and financing decisions. From this perspective, our research not only contributes to finance literature that explores the impact of managerial characteristics on corporate financial decisions, but also extends the upper echelon theory by establishing how corporate behaviors are linked with CEOs' career backgrounds.

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Appendix: Variable Definitions

Variables	Definitions
CEO-related Variables	
<i>#danwei</i>	the number of <i>danwei</i> a CEO has worked for throughout his/her career reflected in his curriculum vitae. We include the focal firm where he serves as CEO and other concurrent positions in other <i>danwei</i> (e.g. board directors at other corporations) but count the different <i>danwei</i> in the same business conglomerate or with similar administrative functions as one <i>danwei</i> .
<i>#danwei2</i>	the number of <i>danwei</i> a CEO has worked for throughout his career reflected in his curriculum vitae. We include the focal firm where he serves as CEO and other concurrent positions in other <i>danwei</i> (e.g. board directors at other corporations), and add the number of <i>danwei</i> regardless of whether they were in the same conglomerate or had the similar administrative functions.
<i>D_#danwei</i>	an indicator variable that is one if <i>#danwei</i> is greater than three and zero otherwise.
<i>D2_#danwei</i>	an indicator variable that is one if <i>#danwei</i> is greater than two and zero otherwise.
<i>D_Ind</i>	an indicator variable that equals one if the CEO has gained experience in more than one industry and zero otherwise. We count the number of industries using the Shenwan Level I industry classification as criteria.
<i>D_Gov</i>	an indicator variable that equals one if the CEO is currently or formerly a government official and zero otherwise.
<i>D_Fin</i>	an indicator variable that equals one if the CEO previously worked in financial institutions including commercial banks, security companies, and trust institutions.
<i>D_NBS</i>	an indicator variable which equals one if the CEO held positions in research institutions and zero otherwise.
<i>Log Age</i>	the log of a CEO's age.
<i>MBA</i>	an indicator variable that equals one if the CEO holds an MBA degree and zero otherwise.
Accounting Variables	
<i>Investment</i>	cash payments for fixed assets, intangible assets, and other long-term assets from the cash flow statement minus cash receipts from selling these assets, scaled by beginning-of-year total assets.
<i>Cash flow</i>	earnings before interest, tax, depreciation, and amortization (EBITDA), scaled by beginning-of-year total assets
<i>Tobin Q</i>	the market value of owner equity and the book value of total liabilities, all divided by the book value of total assets. The market value of tradable shares is calculated based on the year-end price in the stock markets. For non-tradable shares, we set the market price at the book value.
<i>Sales</i>	net sales scaled by beginning-of-year total assets.
<i>Sales growth</i>	current year's sales divided by last year's sales minus one.

<i>Leverage</i>	the ratio of a firm's total liabilities to beginning-of-year total assets.
<i>Bank loans</i>	the ratio of the proceeds from bank loans to the book value of total assets.
<i>Trade credit</i>	the sum of accounts payables and other payables normalized by total assets.
<i>ROA</i>	earnings before interest, tax, depreciation, and amortization (EBITDA) scaled by total assets.
<i>Total assets</i>	the log of total assets in billions of yuan.
Corporate governance measures	
<i>Block</i>	the percentage of shares held by the largest ultimate controlling shareholder of the firm's total outstanding shares.
<i>D_Block</i>	a dummy variable for <i>Block</i> , which is equal to one if the value of <i>Block</i> is larger than the median of the sample and zero otherwise.
<i>Shr2_10</i>	the summation of the square of the ownership of the second and tenth largest shareholders of a company.
<i>D_Shr2_10</i>	a dummy variable for <i>Share2_10</i> that is equal to one if the value of <i>Share2_10</i> is larger than the median of the sample and zero otherwise.
<i>D_CEOshr</i>	an indicator that equals one if the CEO has ownership of the firm and zero otherwise.
Firm characteristics in the year prior to CEO turnover	
<i>Prior ROA</i>	firm's <i>ROA</i> in the year prior to CEO turnover.
<i>Prior Sales growth</i>	firm's <i>Sales growth</i> in the year prior to CEO turnover.
Board and ownership characteristics in the year prior to CEO turnover	
<i>Prior IndDir</i>	independent director percentage of board in the year prior to CEO turnover.
<i>Prior Block</i>	the stock holdings percentage of block shareholder in the year prior to CEO turnover.
Prior CEO characteristics	
<i>Prior CEO tenure</i>	number of years prior CEO had been CEO.
<i>Prior forced turnover</i>	an indicator that equals one if prior CEO left office for abnormal reasons and zero otherwise.

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Table I Summary Statistics

Table I presents summary statistics for all the variables used in our analysis. We use all the CEOs of the firms in the list of the ZhongZheng 800 index (a popular index in China's A share market) at the end of 2010 as the initial sample and exclude the CEOs of financial firms and firms that became public after 2004. The final CEO sample consists of 1332 CEOs. Panel A presents summary statistics for CEO characteristics of 1332 CEOs. Panel B provides frequency statistics of key measures (*#danwei* and *#danwei2*) for the diversity of CEOs' career experiences. Panel C describes the Spearman correlations between CEO characteristics. Finally, we merge the CEO data set with the firm-level financial and accounting variables from 2000 to 2010. Panel D presents the summary statistics for firm characteristics. All continuous variables are winsorised at the 1st and 99th percentiles. Please see the Appendix for detailed definitions of all the variables in Table I.

Panel A: Summary Statistics of CEOs' Characteristics

VARIABLES	N	Mean	Median	Std	Min	Max
<i>#danwei</i>	1332	2.8	2	1.8	1	11
<i>#danwei2</i>	1332	4	4	2	1	16
<i>D_Ind</i>	1332	38%				
<i>D_NBS</i>	1332	16%				
<i>D_Gov</i>	1332	17%				
<i>D_Fin</i>	1332	8%				
<i>MBA</i>	1332	14%				
<i>Outside</i>	1332	34%				
<i>Age</i>	1332	45.3	45	6.7	29	68

Panel B: Frequency Statistics of *#danwei* and *#danwei2*

<i>#danwei</i>	Freq	Pct(%)	Cum Pct(%)	<i>#danwei2</i>	Freq	Pct(%)	Cum Pct(%)
1	438	33	33	1	98	7	7
2	256	19	52	2	199	15	22
3	239	18	70	3	293	22	44
4	171	13	83	4	268	20	64
5	116	9	92	5	200	15	79
6	54	4	96	6	119	9	88
7	36	3	98	7	82	6	95
8 & Above	22	2	100	8	41	3	98
				9	19	1	99
				10 & Above	13	1	100

Panel C: Correlation Matrix of CEOs' Characteristics

	D_#danwei	D2_#danwei	D_Ind	D_Gov	D_NBS	D_FIN	OUTSIDE	MBA	LOGAGE
D_#danwei	1.000								
D2_#danwei	0.674 <.0001	1.000							
D_Ind	0.418 <.0001	0.473 <.0001	1.000						
D_Gov	0.378 <.0001	0.387 <.0001	0.378 <.0001	1.000					
D_NBS	0.190 <.0001	0.205 <.0001	0.234 <.0001	0.093 <.0001	1.000				
D_FIN	0.054 <.0001	0.126 <.0001	0.261 <.0001	0.128 <.0001	-0.023 0.133	1.000			
OUTSIDE	0.153 <.0001	0.182 <.0001	0.188 <.0001	0.090 <.0001	0.054 0.000	0.095 <.0001	1.000		
MBA	0.032 0.032	0.005 0.743	0.022 0.142	-0.012 0.437	0.040 0.008	0.072 <.0001	0.013 0.403	1.000	
LOGAGE	0.010 0.507	0.049 0.001	-0.048 0.001	0.085 <.0001	0.027 0.078	-0.151 <.0001	-0.050 0.001	-0.119 <.0001	1.000

Panel D: Summary Statistics of Firm Characteristics

VARIABLES	N	Mean	Median	Std	Min	Max
Investment	4572	0.08	0.05	0.09	0.00	0.60
Cash flow	4572	0.07	0.07	0.11	-0.38	0.63
Tobin Q	4572	1.50	1.22	0.84	0.76	7.06
Sales	4535	0.65	0.55	0.38	0.05	0.86
Sales growth	4535	0.49	0.10	1.95	-1.12	10.16
Tangibility	4572	0.50	0.51	0.12	0.14	0.91
Leverage	4572	0.52	0.51	0.27	0.05	0.92
ROA	4572	0.04	0.04	0.07	-0.31	0.30
Total Asset(Log)	4572	7.07	3.20	14.27	0.18	48.16
Block	4396	0.41	0.40	0.17	0.04	0.85
Shr2_10	4396	0.02	0.02	0.02	0.00	0.14

Table II Comparisons of Firm Characteristics for Firms with Different Types of CEO

Table II reports the mean and median of corporate characteristics, respectively, for firms with diversely experienced CEOs and those with simply experienced CEOs. Simply experienced CEOs are defined as those who worked for no more than three danwei before he/she joined the current firm, while diversely experienced CEOs refer to those who worked for no less than four danwei. The significance of the mean/median difference between these two groups of firms is also reported. Please see the Appendix for detailed definitions of all the variables in Table II.

<i>Variables</i>	Diversely experienced CEOs: #danwei≤3		Simply experienced CEOs: #danwei>=4		Diversely-Simply	
	Mean	Median	Mean	Median	Dif.Mean	Dif.Median
<i>Investment</i>	0.09	0.05	0.08	0.05	0.01*	
<i>Cash flow</i>	0.07	0.06	0.06	0.06	0.01*	
<i>Tobin Q</i>	1.46	1.18	1.57	1.26	-0.12***	***
<i>Sales growth</i>	0.51	0.10	0.67	0.12	-0.15***	***
<i>Tangibility</i>	0.51	0.51	0.50	0.50	0.01	
<i>Leverage</i>	0.51	0.52	0.53	0.48	-0.02*	**
<i>ROA</i>	0.04	0.04	0.04	0.04	0.00	
<i>Log(Total asset)</i>	9.56	3.79	6.49	3.69	3.07***	
<i>SOE</i>	0.67	1.00	0.51	1.00	0.16***	***
<i>Block</i>	0.44	0.44	0.39	0.37	0.05**	***
<i>Shr2_10</i>	0.02	0.02	0.02	0.01	0.00*	**
<i>Credit Loans Ratios</i>	0.32	0.32	0.31	0.31	0.01**	*
<i>Prior CEO tenure</i>	3.26	2.92	3.15	2.73	0.12*	**
<i>Prior forced turnover</i>	0.22	0.00	0.30	0.00	-0.09***	*
<i>Prior ROA</i>	0.05	0.04	0.02	0.03	0.03***	***
<i>Prior Sales Growth(%)</i>	0.47	0.09	6.06	0.10	-5.59***	
<i>Prior IndDir</i>	0.28	0.27	0.29	0.27	0.01	
<i>Prior Block</i>	0.49	0.51	0.42	0.40	0.07***	***

Table III Investment-Cash Flow Sensitivity and CEOs' Career Experiences: Baseline Results

The dependent variable in the regression is *Investment*, defined as cash payments for fixed assets, intangible assets, and other long-term assets from the cash flow statement minus cash receipts from selling these assets, scaled by beginning-of-year total assets. Panel A presents the regression results using different ways to measure the overall diversity of one's career experiences, while Panel B displays the regression results that explore the impacts of different career experiences (CE) including *D_Ind*, *D_Gov*, *D_Fin*, and *D_NBS*. The definitions of all career experiences measures and other variables in the regression are provided in the Appendix. All standard errors are clustered by firm. Robust t-statistics are reported in parentheses. Here, *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

Panel A

VARIABLES	(1) #danwei	(2) #danwei2	(3) D_#danwei	(4) D2_#danwei	(5) #danwei	(6) #danwei2	(7) D_#danwei	(8) D2_#danwei
<i>Cash Flow</i>	0.217*** (5.04)	0.229*** (4.61)	0.210*** (5.21)	0.212*** (4.80)	0.124*** (4.73)	0.156*** (5.10)	0.105*** (4.99)	0.114*** (5.07)
<i>Tobin Q</i>	0.00780* (1.90)	0.00741* (1.83)	0.00765* (1.86)	0.00803* (1.96)	0.0160*** (6.84)	0.0161*** (6.90)	0.0160*** (6.86)	0.0160*** (6.86)
<i>Tobin Q*Cash flow</i>	-0.0122 (-0.45)	-0.0120 (-0.45)	-0.0121 (-0.46)	-0.0126 (-0.47)	0.00674 (0.80)	0.00696 (0.83)	0.00589 (0.70)	0.00689 (0.82)
<i>Diversity</i>	-0.00604 (-0.98)	-0.000612 (-0.74)	-0.00580 (-0.61)	-0.0107 (-1.12)	-0.00192 (-1.42)	-0.00104 (-0.89)	-0.00949* (-1.91)	-0.00278 (-0.57)
<i>Diversity *Cash Flow</i>	-0.00904** (-2.03)	-0.00905** (-2.06)	-0.0581** (-2.50)	-0.0591** (-2.09)	-0.0147** (-2.27)	-0.0186*** (-3.09)	-0.0698*** (-2.74)	-0.0655*** (-2.77)
	Industry Dummy, Industry Dummy*Cash Flow, Year Dummy, Year Dummy*Cash Flow				Firm Dummy, Firm Dummy*Cash Flow, Year Dummy, Year Dummy*Cash Flow			
<i>Observations</i>	4521	4521	4521	4521	4521	4521	4521	4521
<i>Adj R-squared</i>	0.151	0.148	0.150	0.152	0.316	0.317	0.317	0.316

Panel B

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
VARIABLES	<i>D_Ind</i>	<i>D_Gov</i>	<i>D_NBS</i>	<i>D_Fin</i>	<i>D_Ind</i>	<i>D_Gov</i>	<i>D_NBS</i>	<i>D_Fin</i>
<i>Cash Flow (CF)</i>	0.103*** (4.53)	0.112*** (5.23)	0.119*** (5.52)	0.100*** (4.42)	0.107*** (4.86)	0.107*** (5.06)	0.116*** (5.43)	0.102*** (4.60)
<i>Tobin Q</i>	0.0160*** (6.87)	0.0161*** (6.88)	0.0158*** (6.77)	0.0157*** (6.70)	0.0160*** (6.88)	0.0159*** (6.84)	0.0158*** (6.77)	0.0157*** (6.73)
<i>Tobin Q*CF</i>	0.00649 (0.76)	0.00572 (0.68)	0.00642 (0.76)	0.00681 (0.81)	0.00578 (0.68)	0.00644 (0.76)	0.00636 (0.76)	0.00674 (0.80)
<i>D_#danwei</i>					-0.00714 (-1.35)	-0.0115** (-2.18)	-0.00947* (-1.88)	-0.00829* (-1.66)
<i>D_#danwei*CF</i>					-0.0694** (-2.53)	-0.0560** (-2.01)	-0.0567** (-2.20)	-0.0734*** (-2.80)
<i>CE</i>	-0.0163*** (-2.69)	-0.00705 (-1.00)	-0.0110 (-1.44)	-0.0257*** (-3.48)	-0.00728 (-1.40)	0.00669 (1.06)	-0.00154 (-0.26)	-0.0152*** (-3.01)
<i>CE*CF</i>	-0.0798*** (-2.61)	-0.0800** (-2.21)	-0.131*** (-2.98)	-0.0745** (-2.14)	-0.00368 (-0.14)	-0.0403 (-1.19)	-0.0997*** (-3.08)	-0.0535 (-0.81)
<i>Constant</i>	0.0516*** (4.70)	0.0489*** (4.47)	0.0513*** (4.71)	0.0526*** (4.82)	0.0528*** (4.82)	0.0501*** (4.59)	0.0514*** (4.72)	0.0528*** (4.85)
	Including Firm Dummy, Firm Dummy*Cash Flow, Year Dummy, Year Dummy*Cash Flow							
<i>Observations</i>	4521	4521	4521	4521	4521	4521	4521	4521
<i>Adjusted R-squared</i>	0.329	0.328	0.330	0.329	0.328	0.328	0.330	0.329

Table IV Investment-Cash Flow Sensitivity and CEOs' Career Experiences with All Control Variables

Table IV reports the OLS regression results controlling for the impact of investment opportunities, corporate governance factors and other CEOs' characteristics. The dependent variable in the regression is *Investment*, defined as cash payments for fixed assets, intangible assets, and other long-term assets from the cash flow statement minus cash receipts from selling these assets, scaled by beginning-of-year total assets. In Column (1), we only include the diversity measure (*D_#danwei*) in the regression. *D_Ind*, *D_Gov*, *D_Fin* and *D_NBS* are all measures for CEO's career experiences (CE). In Columns (2) to (5), we include each of these variables in the regression, respectively. To avoid the collinearity problem, we only include *D_Block* and its interaction with cash flow to control for the impact of corporate governance. The definitions of all the variables in the regression are provided in the Appendix. All standard errors are clustered by firm. Robust t-statistics are reported in parentheses. Here *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

VARIABLES	(1)	(2) <i>D_Ind</i>	(3) <i>D_Gov</i>	(4) <i>D_NBS</i>	(5) <i>D_Fin</i>
<i>Cash Flow(CF)</i>	0.114*** (2.95)	0.128*** (3.20)	0.121*** (3.11)	0.135*** (3.40)	0.108*** (2.70)
<i>Tobin Q</i>	0.0208*** (8.39)	0.0208*** (8.39)	0.0209*** (8.43)	0.0207*** (8.34)	0.0205*** (8.26)
<i>Tobin Q*CF</i>	-0.0236*** (-2.66)	-0.0234*** (-2.61)	-0.0242*** (-2.71)	-0.0226** (-2.54)	-0.0225** (-2.53)
<i>D_#danwei</i>	-0.00928* (-1.84)				
<i>D_#danwei*CF</i>	-0.0683*** (-2.61)				
<i>CE</i>		-0.00356 (-0.61)	-0.0101 (-1.40)	0.00283 (0.47)	-0.00949* (-1.91)
<i>CE*CF</i>		-0.0886*** (-2.76)	-0.0605* (-1.65)	-0.0868** (-1.97)	-0.0673* (-1.89)
<i>Sales Growth</i>	0.000141 (0.20)	0.000142 (0.20)	0.000160 (0.22)	0.000115 (0.16)	0.000218 (0.30)
<i>Sales Growth*CF</i>	0.00957*** (2.98)	0.00986*** (3.07)	0.00956*** (2.95)	0.00950*** (2.96)	0.00982*** (3.06)
<i>Leverage</i>	0.0324** (2.40)	0.0337** (2.50)	0.0325** (2.41)	0.0329** (2.44)	0.0350*** (2.59)
<i>Log(Total Asset)</i>	0.0366*** (11.75)	0.0365*** (11.70)	0.0365*** (11.71)	0.0364*** (11.70)	0.0364*** (11.66)
<i>ROA</i>	0.139*** (5.20)	0.141*** (5.27)	0.138*** (5.15)	0.141*** (5.27)	0.140*** (5.24)
<i>Non SOE</i>	0.00543 (1.20)	0.00611 (1.35)	0.00553 (1.22)	0.00543 (1.20)	0.00526 (1.16)
<i>Non SOE*CF</i>	-0.0834*** (-3.53)	-0.0847*** (-3.58)	-0.0842*** (-3.56)	-0.0839*** (-3.55)	-0.0823*** (-3.46)

<i>D_Block</i>	0.0118 (0.66)	0.0126 (0.70)	0.0119 (0.66)	0.0124 (0.69)	0.0101 (0.56)
<i>D_Block*CF</i>	0.00476 (0.07)	0.00258 (0.04)	0.00157 (0.02)	-0.0181 (-0.26)	0.00472 (0.07)
<i>Outside</i>	-0.00596 (-1.34)	-0.00559 (-1.23)	-0.00593 (-1.33)	-0.00524 (-1.17)	-0.00488 (-1.09)
<i>Outside*CF</i>	0.0685*** (2.81)	0.0738*** (2.94)	0.0690*** (2.81)	0.0669*** (2.74)	0.0674*** (2.76)
<i>MBA</i>	-0.0115* (-1.89)	-0.0124** (-2.03)	-0.0123** (-2.01)	-0.0105* (-1.73)	-0.0112* (-1.82)
<i>MBA*CF</i>	0.121*** (3.90)	0.122*** (3.90)	0.125*** (3.98)	0.116*** (3.73)	0.122*** (3.93)
<i>Constant</i>	-0.753*** (-11.50)	-0.754*** (-11.50)	-0.753*** (-11.50)	-0.749*** (-11.44)	-0.746*** (-11.37)
Including Firm Dummy, Firm Dummy*Cash Flow, Year Dummy, Year Dummy*Cash Flow					
<i>Observations</i>	4,396	4,396	4,396	4,396	4,396
<i>Adjusted R-squared</i>	0.377	0.378	0.377	0.378	0.378

Table V Regression of Investment on Cash Flow and CEOs' Career Experiences by Financial Constraint

The dependent variable in the regression is *Investment*. Every year, firms are categorized as a “financially constrained” (FC) group and “not financial constrained” (UFC) group according to three financial constraint indexes described in Section 2.2. We include all control variables in Table IV and run a panel regression in each group. Empirical results are reported in Columns (1) through (3), respectively using *Tangibility*, *Credit loan ratio* and *SOE* as the financial constraint index. For brevity, we report only the value of β_2 to β_5 . *Difference* indicates whether the differences between the two groups are statistically significant, which is determined by the simulation procedure described in Section 4. The null hypothesis is that the coefficients are equal for the two groups under consideration, while the alternative hypothesis is that the coefficient for the UFC group is greater than that of the FC group. The definitions of all the variables in the regression are provided in the Appendix. All regressions include industry/year fixed effects and their interactions with cash flow. All standard errors are clustered by firm. Robust t-statistics are reported in parentheses. Here, *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

	(1)		(2)		(3)	
	<i>KZ index</i>		<i>Tangibility</i>		<i>Credit Loan Ratio</i>	
VARIABLES	FC	UFC	FC	UFC	FC	UFC
<i>Cash Flow</i>	0.153*** (2.67)	0.0977 (1.47)	0.260*** (4.70)	-0.0187 (-0.61)	0.174*** (7.61)	0.0921** (2.07)
<i>Tobin Q</i>	0.0174*** (3.30)	0.0222*** (6.99)	0.0254*** (5.59)	0.0118*** (4.12)	0.0194*** (7.38)	0.0112*** (2.68)
<i>D_#danwei</i>	0.00382 (0.40)	-0.0238*** (-3.11)	-0.0130 (-1.36)	-0.00563 (-0.83)	-0.0116* (-1.89)	-0.00436 (-0.47)
<i>D_#danwei *Cash flow</i>	-0.0690* (-1.96)	-0.0325 (-0.85)	-0.210*** (-3.54)	-0.0386 (-1.22)	-0.0896*** (-2.88)	-0.0289 (-0.60)
<i>Difference (P-Value)</i>	0.086*		0.013**		0.042**	

Table VI CEOs' Career Experiences and Firms' Outside Financing

Table VI reports the impact of CEOs' career experiences on firms' outside financing. Panel A displays the results using bank loans as the dependent variable, while Panel B shows the results using trade credit as the dependent variable. Firm and year fixed effects are controlled in all regressions. *Bank loans* are defined as the ratio of the proceeds from bank loans to the book value of total assets, while *trade credit* is the sum of accounts payables and other payables normalized by total assets. In Column (1), we only include the diversity measure (*D_#danwei*) in the regression. *D_Ind*, *D_Gov*, *D_Fin* and *D_NBS* are all measures for CEOs' career experiences (CE). In Columns (2) to (5), we include each of these variables in the regression, respectively. Control variables are documented in Section 5. The definitions of these CEOs' career experiences measures and other control variables in the regression are provided in the Appendix. Panel C shows the OLS regression results with industry fixed effects among firms with different financial constraints. Every year, firms are categorized as a "financially constrained" (FC) group and "not financially constrained" (UFC) group according to three financial constraint indexes described in Section 2.2. We include all control variables in Panel A. To save space, we use the sum of bank loans and trade credit as dependent variables and only report the estimates of the diversity measure (*D_#danwei*). All standard errors are clustered by firm. Robust t-statistics are reported in parentheses. Here, *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

Panel A: Bank Loans					
	(1)	(2)	(3)	(4)	(5)
VARIABLES		<i>D_Ind</i>	<i>D_Gov</i>	<i>D_NBS</i>	<i>D_Fin</i>
<i>D_#danwei</i>	0.0152*** (2.59)				
<i>Career Experiences</i>		0.0256*** (3.51)	0.0226*** (2.68)	0.0134 (1.46)	0.0329*** (3.60)
<i>Sales</i>	-7.745 (-1.18)	-7.305 (-1.11)	-7.935 (-1.20)	-7.765 (-1.18)	-7.085 (-1.08)
<i>ROA</i>	-0.491*** (-16.43)	-0.488*** (-16.34)	-0.490*** (-16.40)	-0.491*** (-16.42)	-0.490*** (-16.41)
<i>Tobin Q</i>	8.095 (1.33)	8.155 (1.34)	8.455 (1.39)	8.095 (1.33)	7.925 (1.30)
<i>Capital Expenditures</i>	-0.0424** (-2.04)	-0.0414** (-1.99)	-0.0424** (-2.04)	-0.0426** (-2.05)	-0.0398* (-1.92)
<i>Tangibility</i>	0.165*** (8.37)	0.165*** (8.37)	0.164*** (8.31)	0.165*** (8.36)	0.163*** (8.28)
<i>Log(age)</i>	0.0110 (0.67)	0.0129 (0.79)	0.0114 (0.69)	0.0113 (0.69)	0.0209 (1.25)
<i>D_CEOshr</i>	-0.0151*** (-2.58)	-0.0147** (-2.52)	-0.0154*** (-2.64)	-0.0151*** (-2.58)	-0.0148** (-2.54)
<i>Constant</i>	0.236*** (3.68)	0.225*** (3.51)	0.235*** (3.67)	0.235*** (3.66)	0.197*** (3.01)
<i>Year Dummy</i>	Yes	Yes	Yes	Yes	Yes
<i>Firm Dummy</i>	Yes	Yes	Yes	Yes	Yes

<i>Observations</i>	4,341	4,341	4,341	4,341	4,341
<i>Adjusted R-squared</i>	0.543	0.543	0.543	0.543	0.544
Panel B: Trade Credit					
	(1)	(2)	(3)	(4)	(5)
VARIABLES		D_Ind	D_Gov	D_NBS	D_Fin
<i>D_#danwei</i>	0.00882** (1.97)				
<i>Career Experiences</i>		0.0180** (2.43)	0.00429 (0.50)	0.0119* (1.77)	0.0271*** (2.91)
<i>Sales</i>	-3.946 (-0.16)	1.467 (0.12)	-2.886 (-0.14)	-2.836 (-0.24)	2.886 (0.14)
<i>ROA</i>	-0.647*** (-21.26)	-0.644*** (-21.16)	-0.648*** (-21.27)	-0.648*** (-21.29)	-0.646*** (-21.25)
<i>Tobin Q</i>	4.085 (0.66)	4.325 (0.70)	3.875 (0.62)	4.005 (0.65)	3.905 (0.63)
<i>Capital Expenditures</i>	0.136*** (6.43)	0.136*** (6.42)	0.136*** (6.43)	0.137*** (6.47)	0.139*** (6.56)
<i>Tangibility</i>	0.101*** (5.02)	0.100*** (5.00)	0.101*** (5.03)	0.101*** (5.03)	0.0988*** (4.92)
<i>Log(age)</i>	-0.0130 (-0.78)	-0.0117 (-0.70)	-0.0132 (-0.79)	-0.0137 (-0.82)	-0.00274 (-0.16)
<i>D_CEOshr</i>	-0.00745 (-1.25)	-0.00700 (-1.18)	-0.00725 (-1.22)	-0.00740 (-1.24)	-0.00720 (-1.21)
<i>Constant</i>	0.414*** (6.36)	0.404*** (6.19)	0.415*** (6.36)	0.415*** (6.35)	0.374*** (5.62)
<i>Year Dummy</i>	Yes	Yes	Yes	Yes	Yes
<i>Firm Dummy</i>	Yes	Yes	Yes	Yes	Yes
<i>Observations</i>	4,341	4,341	4,341	4,341	4,341
<i>Adjusted R-squared</i>	0.442	0.442	0.442	0.442	0.442

Panel C Regressions among Firms with Different Financial Constraints						
	(1)		(2)		(3)	
	KZ index		Tangibility		Credit Loan Ratio	
	FC	UFC	FC	UFC	FC	UFC
<i>D_#danwei</i>	0.0197** (2.49)	-0.00856 (-0.66)	0.0184** (2.03)	0.0113 (1.38)	0.0303*** (3.04)	0.0161 (1.10)
Difference (P-Value)	0.069*		0.172		0.041**	

Table VII Heckman Two-Stage Procedures

Table VII reports the results of the Heckman two-step treatment effect model. Panel A reports the results of the first-stage probit regression. The dependent variable is the diversity of CEOs' career experiences (*D_#danwei*) and explanatory variables include *Prior CEO tenure*, *Prior forced turnover*, *Prior ROA*, *Prior sales growth*, *Prior independent directors' ratio*, *Prior block shareholder ownership*, industry dummy variables, and other variables in the second stage regressions. The likelihood that a firm would hire a well-connected CEO (*Lambda*), as obtained from a first-stage probit regression, is included as a control variable in the second-stage analysis. Panel B reports the second stage results using *Investment* as the dependent variable. Other variables in Table IV are also included in Panel B. For brevity, we only report the estimated value of β_2 to β_5 and *Lambda*. Panel C reports the second stage results using the sum of *bank loans* and *trade credit* as the dependent variable. The definitions of all the variables are provided in the Appendix. All standard errors are clustered by firm. Robust t-statistics are reported in parentheses. Here, *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

VARIABLES	(1)
Panel A: First-Stage Probit Model of the Heckman Test	
<i>Prior CEO tenure</i>	-0.0485*** (-3.77)
<i>Prior forced turnover(0/1)</i>	0.168** (2.57)
<i>Prior ROA</i>	-0.896*** (-3.54)
<i>Prior Sales Growth</i>	0.00133** (2.27)
<i>Prior IndDir</i>	0.384 (1.24)
<i>Prior Block</i>	-0.729*** (-4.14)
<i>Year Dummy</i>	Yes
<i>Industry dummy</i>	Yes
Panel B: Second-Stage Regression: Investment-Cash Flow Sensitivity	
<i>Cash flow</i>	0.122*** (3.53)
<i>Tobin Q</i>	0.0216*** (7.79)
<i>D_#danwei</i>	-0.0368 (-1.48)
<i>D_#danwei *Cash flow</i>	-0.0702** (-2.18)
<i>Lambda</i>	0.0193 (1.35)
Panel C: Second-Stage Regression: Bank Loans + Trade Credit	
<i>D_#danwei</i>	0.013** (2.11)

<i>Sales</i>	0.000245 (1.50)
<i>ROA</i>	-1.407*** (-16.25)
<i>Tobin Q</i>	-0.00925 (-1.16)
<i>Capital Expenditures</i>	0.234*** (3.50)
<i>Tangibility</i>	0.456*** (9.88)
<i>Log(age)</i>	-0.0176 (-0.47)
<i>D_CEOshr</i>	-0.00152 (-0.13)
<i>Lambda</i>	-0.0822** (-2.23)
<i>Year dummy</i>	Yes
<i>Industry dummy</i>	Yes
<i>Observations</i>	3,293

Table VIII CEOs' Career Experiences and Exogenous Shocks on Outside Financing

The dependent variable is *Investment*, defined as cash payments for fixed assets, intangible assets, and other long-term assets from the cash flow statement minus cash receipts from selling these assets, scaled by beginning-of-year total assets. *Cash flow* is defined as earnings before interest, tax, depreciation, and amortization (EBITDA), scaled by beginning-of-year total assets. *Tobin Q* equals the market value of owner equity and the book value of total liabilities, all divided by the book value of total assets. The market value of tradable shares is calculated based on the year-end price in the stock markets. (For non-tradable shares, we set the market price at book value). *Credit_Gr* is defined as the growth rate of the total credit supply. *D_#danwei*, is used to measure the diversity of a CEO's career experiences. Table VIII reports the OLS regression results including firm/year fixed effects and their interactions with *Credit_Gr*. All standard errors are clustered by firm. Robust t-statistics are reported in parentheses. Here, *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

VARIABLES	(1)
<i>Cash flow</i>	0.094*** (4.36)
<i>Tobin Q</i>	0.0171*** (3.71)
<i>D_#danwei</i>	-0.00972** (-1.96)
<i>Credit_Gr</i>	0.0949** (2.56)
<i>Credit_Gr *D_#danwei</i>	-0.0188* (-1.86)
<i>Year fixed effect</i>	Yes
<i>Year fixed effect*Credit_Gr</i>	Yes
<i>Firm fixed effect</i>	Yes
<i>Firm fixed effect*Credit_Gr</i>	Yes
<i>Observations</i>	4,396
<i>Adj R-square</i>	0.39
<i>Number of firms</i>	559

Table IX The Instrumental Regressions

Table IX reports regression results with the corporate headquarters location (Capital) as an instrumental variable for the diversity of CEOs' career experiences. Capital is a dummy variable, which equals one when the city where a firm headquartered is a vice provincial city and zero otherwise. The column headed "1st Stage" reports the regression results in the first stage. The column headed "2nd Stage" reports the regression results in the second stage. Panel A reports the results of investment-cash flow sensitivity. Panel B reports the results for external debt financing. In addition to all control variables in Table IV and Table VI, we add the per capita GDP of the city where a firm is headquartered and its interactions with cash flow into the regression. The definitions of all the variables are provided in the Appendix. All standard errors are clustered by firm. Robust t-statistics are reported in parentheses. Here, *, **, and *** indicate the significance at the 10%, 5%, and 1% levels, respectively.

Panel A

	(1)	(2)	(3)
	Dependent Variables		
	<i>1st Stage</i>	<i>2nd Stage</i>	
VARIABLES	<i>D_#danwei</i>	<i>D_#danwei*Cash Flow</i>	<i>Investment</i>
<i>D_#danwei</i>			-0.0872 (-1.57)
<i>D_#danwei*Cash Flow</i>			-0.670** (-2.48)
<i>Capital</i>	0.0459*** (3.06)	-0.0018*** (-2.90)	
<i>Capital*Cash Flow</i>	0.138 (1.28)	0.0951*** (6.57)	
<i>Cash Flow</i>	-0.850 (-0.83)	-0.493*** (-3.59)	0.414* (1.91)
<i>Tobin Q</i>	0.0152 (1.55)	-0.00361*** (-2.77)	0.0149*** (4.83)
<i>Tobin Q*Cash Flow</i>	-0.0125 (-0.32)	0.0221*** (4.20)	-0.0248** (-2.08)
<i>Sales Growth</i>	-0.00261 (-0.90)	-0.000535 (-1.38)	-0.000929 (-1.15)
<i>Sales Growth*Cash Flow</i>	-0.00370 (-0.27)	-0.00335* (-1.80)	0.00516 (1.34)
<i>Leverage</i>	-0.0172 (-0.71)	-0.000445 (-0.14)	0.0114* (1.67)
<i>Log(Total Asset)</i>	-0.0362*** (-5.91)	-0.00343*** (-4.20)	0.0113*** (4.91)
<i>ROA</i>	0.149 (1.42)	0.0273* (1.95)	0.156*** (5.35)
<i>Non SOE</i>	0.0274* (1.90)	-0.000131 (-0.07)	0.00709* (1.70)

<i>Non SOE*Cash Flow</i>	-0.0997 (-0.99)	-0.00851 (-0.63)	-0.109*** (-3.92)
<i>Block</i>	-0.156*** (-3.79)	0.000652 (0.12)	-0.00499 (-0.36)
<i>Block*Cash Flow</i>	0.305 (1.11)	-0.165*** (-4.50)	-0.219** (-2.31)
<i>Outside</i>	0.0525*** (3.80)	-0.000464 (-0.25)	-0.00388 (-0.80)
<i>Outside*Cash Flow</i>	-0.128 (-1.30)	0.00244 (0.19)	0.0299 (1.07)
<i>MBA</i>	0.0487*** (2.71)	0.00898*** (3.75)	0.0148*** (2.69)
<i>MBA*Cash Flow</i>	-0.0859 (-0.69)	-0.0820*** (-4.91)	0.0885** (2.27)
<i>Per Capita GDP</i>	0.0619*** (3.35)	0.000330 (0.13)	-0.000529 (-0.16)
<i>Per Capita GDP*Cash Flow</i>	0.0573 (0.53)	0.0684*** (4.76)	-0.00462 (-0.42)
<i>Constant</i>	0.421** (2.07)	0.0698** (2.57)	-0.190*** (-3.20)
<i>Year fixed effect</i>	Yes	Yes	Yes
<i>Year fixed effect*Cash flow</i>	Yes	Yes	Yes
<i>Industry fixed effect</i>	Yes	Yes	Yes
<i>Industry fixed effect*Cash flow</i>	Yes	Yes	Yes
<i>F-Statistics (Excluded IVs)</i>	10.625	25.321	
<i>Partial R-sqr</i>	0.009	0.021	
<i>Observations</i>	4396	4396	4396
<i>Adj R-squared</i>	0.161	0.481	0.323

Panel B

<i>VARIABLES</i>	(1) 1st Stage <i>D_#danwei</i>	(2) 2nd Stage <i>Leverage</i>
<i>D_#danwei</i>		0.473** (2.31)
<i>Capital</i>	0.0406*** (3.21)	
<i>Sales</i>	-0.000109 (-0.47)	0.000217 (1.17)
<i>ROA</i>	0.0377 (0.43)	-1.396*** (-19.83)

<i>Tobin Q</i>	-6.98e-05 (-0.33)	0.00139*** (8.24)
<i>Capital Expenditures</i>	-0.263*** (-4.14)	0.290*** (3.86)
<i>Tangibility</i>	-0.0532 (-1.05)	0.196*** (4.64)
<i>Log(age)</i>	0.0899** (2.31)	-0.138*** (-3.78)
<i>D_CEOshr</i>	-0.0223* (-1.87)	0.000437 (0.04)
<i>Per Capita GDP</i>	0.0407** (2.49)	-0.00183 (-0.20)
<i>Constant</i>	-0.467** (-2.20)	0.747*** (4.83)
<i>Year fixed effect</i>	Yes	Yes
<i>Year fixed effect*Cash flow</i>	Yes	Yes
<i>Industry fixed effect</i>	Yes	Yes
<i>Industry fixed effect*Cash flow</i>	Yes	Yes
<i>F-Statistics (Excluded IVs)</i>	12.21	
<i>Partial R-sqr</i>	0.011	
<i>Observations</i>	4,396	4,396
<i>Adj R-squared</i>	0.084	0.35

Highlights

- We highlight the effect of CEOs' career experiences on corporate investment.
- Firms with diversely experienced CEOs exhibit lower investment-cash flow sensitivity and exploit more outside funds.
- These effects are more pronounced among financially constrained firms.
- The overall diversity is a broad measure to capture CEOs' connections.